

Global Energy Storage

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America

- [LG Energy Solution launches battery production in Lansing.- electrive.com](#) LG Energy Solution (LGES) has started production at its Lansing, Michigan battery plant, a facility with more than 35 GWh of annual capacity, over US\$2 billion in investment, and plans to expand employment from 900 to 1,700 workers. Originally developed as a GM joint venture, the site is now wholly owned by LGES and will produce both NMC batteries for Toyota EVs and LFP batteries for energy storage systems (ESS). The plant will supply cells for Toyota's 2027 electric Highlander and Tesla's Megapack storage systems, backed by a US\$4.3 billion supply agreement. The launch reflects LGES's broader strategy to diversify beyond EVs and expand its North American ESS business, targeting more than 50 GWh of ESS battery capacity in the region by the end of 2026.
- [LG Energy Mulls Supplying Batteries for US Drones.-BNEF](#) LG Energy Solution (LGES) is exploring opportunities to supply batteries for US military drones and unmanned systems, marking a potential expansion into the defense sector as EV demand slows. The company has received inquiries from multiple US partners while simultaneously shifting its North American focus toward energy storage, having repurposed 5 of its 8 regional battery plants for ESS applications. Key growth drivers include a US\$4.3 billion LFP battery supply agreement for Tesla Megapack systems beginning in 2027, a US\$1.5 billion Toyota battery order, and expansion of its Lansing, Michigan plant to 35 GWh annual capacity by the end of 2027. The move reflects growing demand from US energy storage, AI data centers, and defense supply chains amid efforts to reduce reliance on Chinese battery and drone imports.
- [LG, Nvidia to jointly develop humanoid robot for 2027 unveiling.- Korea joongang Daily](#) LG Group and Nvidia have signed a strategic partnership covering humanoid robots, AI factories, and autonomous driving technologies. LG plans to unveil Nvidia-powered humanoid robots by Q1 2027, using Nvidia's Jetson Thor, Isaac GR00T, and Halos for Robotics platforms, while LG will supply actuators, sensors, and batteries through its affiliates. The companies also plan to build an AI factory pilot facility in H1 2027, followed by an 80 MW AI data center in Cheonan, South Korea, in H1 2028. The collaboration extends to AI-defined vehicles, combining Nvidia's autonomous driving platforms with LG's automotive electronics expertise, further strengthening both companies' positions in next-generation AI infrastructure and robotics.

Asia

- [BYD Tries Where Others Failed With Tiny EV Designed for Japan.-BNEF](#) BYD is seeking to expand its presence in Japan with the launch of the Racco, a compact kei-class EV designed specifically for Japan's roads and consumer preferences. The model received more than 1,000 orders within weeks of launch and is priced at around ¥2 million (US\$12,600) after subsidies. BYD is targeting Japan's largest vehicle segment, where more than 1.6 million kei cars were sold in 2025, representing over one-third of total new vehicle sales. Despite challenges from strong domestic brands, subsidy disadvantages, and consumer preference for local automakers, BYD aims to build its brand and grow EV adoption in Japan, where it sold approximately 7,400 vehicles since entering the market in 2023 and 3,742 EVs in 2025, approaching Toyota's 4,227 EV sales in the country.
- [Geely to trial solid-state batteries from 2027.- electrive.com](#) Geely plans to start pilot deployment of its solid-state batteries in 2027, targeting an energy density of up to 500 Wh/kg at the pack level, significantly higher than current EV batteries. The company says the technology could enable driving ranges exceeding 1,000 km on a single charge. Vehicle testing will span brands across the Geely portfolio, including Geely, Zeekr, Volvo, Polestar, Lotus, and Smart, although no specific models have been announced. The move places Geely alongside BYD and CATL, both of which are also targeting 2027 for early solid-state battery pilot production, while large-scale

commercialization is still expected toward the end of the decade.

- [Nio launches first fifth-generation battery swap stations.- electrive.com](#) Nio has launched its first fifth-generation battery swap stations in China and surpassed 4,000 swap stations nationwide, with the milestone site opening in Quanzhou. The new stations support all three Nio brands, including Firefly for the first time, thanks to a redesigned platform that accommodates vehicles with wheelbases up to 3.5 meters. Nio also completed its 120 millionth battery swap, adding 20 million swaps in just six months after reaching 100 million in February. The company now operates 4,006 swap stations, 5,160 charging stations, and 29,801 charging points across China, covering more than 550 cities, with 1,049 highway swap stations and 80.1% of users having a swap station within 3 km of home or work.
- [POSCO to Supply Lithium to China's Ronbay Group.- thelec.net](#) POSCO Holdings has signed a strategic agreement with Chinese cathode materials producer Ronbay Group to supply battery-grade lithium and expand cooperation across the battery materials value chain. POSCO plans to complete qualification of its battery-grade lithium hydroxide from the Yulchon plant in South Korea by Q4 2026, paving the way for commercial supply. The partnership covers hard-rock lithium, brine-based lithium from Argentina, and recycled lithium from POSCO's recycling operations. The companies will also deepen collaboration in battery recycling, circular materials, LFP batteries, and ESS battery materials, supporting POSCO's efforts to expand its presence in China's battery materials market.
- [Shinsung E&G Expands Energy Businesses Through Startup Partnerships.- thelec.net](#) Shinsung E&G is expanding its clean energy and AI infrastructure businesses through partnerships with several startups, focusing on virtual power plants (VPPs), mobile energy storage systems (ESS), and immersion cooling for AI data centers. The company is leveraging 60Hertz's VPP platform to develop power forecasting and electricity trading services, while a pilot project with aeonus demonstrated a mobile ESS capable of transporting renewable energy between sites. In the data center sector, Shinsung E&G plans to integrate DataBean's immersion cooling technology into projects, including a recently secured KRW 10 billion (US\$7 million) data center project in Malaysia. The company is also collaborating with Korea Green Data on AI-based energy management software and investing in climate-tech, battery, and materials startups through venture funds.
- [Tesla's Giga Shanghai rolls out 6 millionth battery pack.- cnevpost](#) Tesla's Giga Shanghai has produced its 6 millionth battery pack, reaching the milestone just 9 months after producing its 5 millionth pack in November 2025. The factory, Tesla's largest global manufacturing site, has an annual vehicle production capacity of about 1 million units and has contributed nearly 50% of Tesla's global EV deliveries since opening. Despite weaker domestic demand, with July China deliveries falling 32.9% YoY to 27,249 vehicles, exports remain strong, reaching a record 66,330 vehicles in July. In the first seven months of 2026, exports rose 130.1% YoY to 295,324 vehicles, exceeding total exports for all of 2025 and helping maintain high utilization rates at the Shanghai plant.
- [CATL Invests RMB 4.1 Billion in Zhongheng Electric's Parent Company.- solare.com](#) CATL will invest approximately RMB 4.1 billion for a 49% stake in Hangzhou Zhongheng Technology Investment Co., Ltd., the controlling shareholder of Zhongheng Electric (Hangzhou Century Co., Ltd.). Beyond the equity investment, the companies have formed a strategic partnership covering AI/data center power infrastructure, EV charging and battery-swapping solutions, and next-generation power systems, including virtual power plants and green electricity trading, while CATL will also gain certain governance rights through board and management nominations.
- [CATL Establishes RMB 6.1bn Energy Services Holding Company in Xiamen.- solare.com](#) CATL has established Shidai Dianfu (Xiamen) Holding Co., Ltd. in Xiamen with a registered capital of RMB 6.1 billion, wholly owned by the company. The new platform will cover battery manufacturing and sales, battery-swapping infrastructure, power equipment, solar energy services, energy storage, and AI software development. The move is part of CATL's broader strategy to build a comprehensive ecosystem in Xiamen, following the establishment of an RMB 30 billion resource investment platform, an RMB 10 billion zero-carbon technology company, and the development of an 80GWh lithium-ion battery manufacturing base, further strengthening its integrated "resources-manufacturing-services-low carbon" value chain.
- [Solid-State Battery Developer Launches RMB 6bn, 30GWh Manufacturing Project in Inner Mongolia.- solare.com](#) Tuoyi Gooneng (Inner Mongolia) Technology Co., Ltd. has started construction of a RMB 6 billion solid-state battery manufacturing project in Hohhot, Inner Mongolia. The project will be developed in three phases, with each phase adding a 10GWh production line, resulting in a total annual capacity of 30GWh upon completion. The first phase includes battery cell production facilities, electrode workshops, R&D centers, and logistics infrastructure. Leveraging next-generation solid-state battery technology, the project aims to serve applications including electric vehicles, energy storage systems, grid balancing, and industrial power supplies, reflecting growing investment in China's emerging solid-state battery sector.
- [Star Sky Group Signs Sodium-Ion Battery Smart Factory Project in Zhejiang.- solare.com](#) Star Sky Group has signed an agreement with the Huzhou Nanxun Economic Development Zone in Zhejiang Province to build a sodium-ion battery smart factory, marking a major step in the company's Yangtze River Delta expansion strategy. The project will focus on the R&D and manufacturing of sodium-ion battery materials, electrolytes, and battery cells, targeting applications such as grid-scale energy storage, commercial and industrial storage, and low-speed electric vehicles. The investment reflects growing momentum behind sodium-ion battery commercialization in China, supported by the

technology's advantages in cost, safety, resource availability, and low-temperature performance.

- [CATL and Dongfeng Liuzhou Motor Sign Five-Year Strategic Partnership.- solare.com](#) CATL and Dongfeng Liuzhou Motor Co., Ltd. (Dongfeng Liuzhou Motor) have signed a five-year strategic cooperation agreement to deepen collaboration across the new energy commercial vehicle value chain. The partnership will focus on jointly developing battery technologies and standards for a wide range of commercial vehicle applications, including trucks, specialty vehicles, and hydrogen-powered models. The two companies will also explore innovative business models such as Battery Bank and battery-as-a-service (BaaS) solutions to reduce vehicle acquisition costs and operating risks, supporting the broader adoption of low-carbon commercial transportation.

Europe

- [Schaeffler and CATL to deepen battery management collaboration.- electrive.com](#) Schaeffler and CATL have signed a cooperation agreement to jointly develop battery management systems (BMS) and integrated "X-in-1" power electronics solutions for electric vehicles, initially targeting the European market. Building on an existing customer project, Schaeffler will contribute expertise in hardware, software, system integration, testing, and European manufacturing, while CATL will provide battery, algorithm, and large-scale production capabilities. The companies will also explore the use of AI and cloud computing in battery management. The partnership supports the localization of Europe's EV supply chain and strengthens CATL's efforts to expand its battery technology footprint among European automakers.
- [Neptune Energy advances Altmark lithium project.- electrive.com](#) Neptune Energy and Fraunhofer IEG have entered the second pilot phase of the Altmark Lithium Extraction (ALE) project in Germany, testing direct lithium extraction (DLE) technologies to support future commercial production. The company aims to begin lithium production in 2030 and eventually reach 25,000 tonnes of lithium carbonate equivalent (LCE) per year, enough to supply batteries for approximately 500,000 EVs annually. The Altmark region contains an estimated 43 million tonnes of LCE resources, making it one of the world's largest known lithium resources linked to a single project. The pilot plant can process 1,000 liters of thermal water per day as Neptune seeks to establish a domestic European lithium supply chain for the battery and EV industry.

Others

- [Trump-Backed Dronemaker Signs Deal With Gulf-Based Energy Firm.- BNEF](#) Powerus, a US drone and counter-drone company backed by Eric Trump and Donald Trump Jr., has secured a US\$22.3 million contract with an undisclosed energy company in the Persian Gulf to provide drone detection and tracking systems. The deal reflects rising demand for private-sector counter-drone protection following recent regional conflicts and increasing threats to critical energy infrastructure. The system will include sensors, automation technology, and operational support, with future compatibility for Powerus' Guardian interceptor platform. The agreement highlights growing investment in drone defense capabilities as companies seek to protect high-value assets from low-cost aerial threats.
- [One in 25 Australian Homes Bought a Battery in Grid Shakeup.- BNEF](#) Australia has surpassed 500,000 residential battery installations, equivalent to about 4% of all households, just 14 months after launching a government subsidy program. Home batteries now provide approximately 14 GWh of storage capacity, exceeding the 9 GWh of residential storage deployed across the entire United States despite Australia's much smaller population. The rapid adoption is helping support the country's renewable energy transition, with utilities increasingly integrating household batteries into virtual power plants (VPPs). Origin Energy's VPP network now connects more than 400,000 homes and businesses and exceeds 1.5 GW of capacity. Installations have significantly outpaced expectations, with over 9 GWh of small-scale storage deployed in the first half of 2026 alone.

EXHIBIT 1: **Key commodities price performance**

	Price 21-Aug	Ccy	Mkt cap \$bn	P/E 26E	EV /EBITDA 26E	%1D	%1W	%1M	%1Y	1Y px chart
Metal prices (US\$/tonne)										
LiCO (spot)	21,934					0.8%	4%	3%	83%	
LiOH (spot)	20,447					0.8%	4%	5%	84%	
LiCO (contract)	19,900					0.0%	0%	0%	112%	
LiOH (contract)	19,400					0.0%	0%	-2%	128%	
Cobalt (spot)	55,855					0.0%	0%	0%	70%	
Nickel (spot)	16,880					-1.4%	0%	-1%	13%	
Component prices										
Cathode NMC811						0.0%	-1%	-2%	34%	
Cathode LFP						0.0%	0%	4%	51%	
Precursor NMC811						0.0%	-4%	-8%	17%	
Precursor LFP						0.0%	0%	3%	50%	
Artificial Graphite						0.0%	0%	0%	6%	
Natural Graphite						0.0%	0%	0%	0%	
Electrolyte NMC						0.0%	0%	5%	36%	
Electrolyte LFP						0.0%	0%	5%	53%	
Separator						0.0%	0%	0%	28%	
Spot battery cell cost (US\$/kWh)										
LFP	60					0.0%	0%	1%	15%	
NMC532	84					0.0%	-1%	-2%	21%	
NMC622	81					0.0%	-1%	-2%	16%	
NMC811	83					0.0%	-1%	-1%	18%	
Spot battery pack cost (US\$/kWh)										
LFP	73					0.0%	0%	-3%	-2%	
NMC532	116					0.0%	-1%	-2%	14%	
NMC622	114					0.0%	-1%	-2%	11%	
NMC811	116					0.0%	0%	0%	13%	

Source: Bloomberg, Baiinfo, Bernstein analysis and estimates (cell and pack cost)

EXHIBIT 2: Key companies price performance and valuation

	Price 21-Aug	Ccy	Mkt cap \$bn	P/E 26E	EV /EBITDA 26E	%1D	%1W	%1M	%1Y	1Y px chart
Cell makers										
LGES	348,000	KRW	59.0	n.a.	15.0x	-2.9%	-6%	9%	-7%	
Samsung SDI	486,000	KRW	28.4	74.5x	15.6x	-2.5%	-6%	16%	126%	
CATL	395.0	CNY	277.2	19.0x	13.3x	2.6%	0%	3%	42%	
CALB	20.3	HKD	4.6	11.6x	3.1x	2.3%	3%	18%	-7%	
EVE Energy	56.3	CNY	18.2	16.8x	10.6x	2.3%	0%	4%	17%	
Gotion	27.1	CNY	7.3	26.6x	6.4x	2.4%	0%	-5%	-13%	
Sunwoda	18.1	CNY	5.0	14.1x	5.1x	1.5%	-1%	5%	-18%	
Farasis	9.8	CNY	1.8	n.a.	n.a.	1.6%	-2%	2%	-45%	
Lithium Miners/Refiners										
Albemarle	134.2	USD	15.8	11.4x	5.6x	-0.1%	-1%	13%	70%	
SQM	78.5	USD	20.8	10.8x	5.4x	4.1%	6%	14%	73%	
Pilbara Minerals	5.1	AUD	11.6	27.8x	14.3x	4.7%	4%	23%	129%	
Ganfeng Lithium	40.7	HKD	15.5	9.6x	8.3x	8.8%	3%	11%	30%	
Tianqi Lithium	48.3	CNY	11.7	12.5x	4.2x	4.9%	-1%	11%	13%	
Cathode										
LG Chem	254,000	KRW	13.0	n.a.	2.4x	-4.0%	-9%	0%	-13%	
EcoPro BM	106,100	KRW	7.5	575.3x	51.9x	-6.4%	-9%	0%	-18%	
L&F	105,000	KRW	3.1	n.a.	14.1x	-2.4%	-4%	25%	21%	
Umicore	22.4	EUR	6.5	12.5x	5.5x	-4.4%	-4%	12%	66%	
Ningbo Ronbay	27.1	CNY	2.9	58.0x	13.0x	3.5%	0%	2%	21%	
Beijing Easpring	41.8	CNY	3.4	20.8x	13.4x	2.6%	0%	2%	-3%	
GEM	6.5	CNY	5.0	16.7x	5.2x	1.1%	-2%	0%	-4%	
Huayou Cobalt	40.7	CNY	11.5	9.7x	4.3x	3.5%	-4%	3%	-10%	
Dynanonic	47.5	CNY	2.0	16.1x	6.3x	4.7%	0%	12%	33%	
Anode										
Posco	155,300	KRW	10.0	406.1x	41.3x	-4.7%	-9%	12%	2%	
Putailai	24.5	CNY	7.8	15.9x	9.9x	3.4%	0%	7%	34%	
Shanshan	11.9	CNY	4.0	n.a.	n.a.	1.7%	-2%	-5%	1%	
Separator										
Asahi Kasei	1,634	JPY	14.0	15.1x	5.5x	1.1%	-4%	-12%	32%	
SK IE	13,860	KRW	0.8	n.a.	-17.0x	-4.1%	-11%	4%	-52%	
Toray Industries	1,292	JPY	12.2	24.0x	7.4x	-0.7%	-5%	11%	32%	
W-Scope	151	JPY	0.1	n.a.	n.a.	-1.3%	-5%	-1%	-39%	
Yunnan Energy	57.6	CNY	8.4	26.8x	10.3x	3.2%	2%	15%	83%	
Shenzhen Senior	14.7	CNY	3.1	30.5x	9.7x	1.0%	-2%	15%	22%	
Sinoma Science	53.5	CNY	13.3	30.1x	12.5x	-0.2%	-11%	0%	67%	
Electrolyte										
Enchem	18,040	KRW	0.3	n.a.	n.a.	-15.1%	-27%	-10%	-78%	
Tinci Minerals	38.4	CNY	11.6	11.7x	8.7x	3.5%	-3%	6%	89%	
Capchem	64.5	CNY	7.2	22.8x	15.8x	-1.1%	-4%	4%	54%	
Jiangsu Guotai	7.7	CNY	1.9	9.8x	n.a.	-0.6%	0%	2%	2%	
Shanshan	11.9	CNY	4.0	n.a.	n.a.	1.7%	-2%	-5%	1%	
Equipment Maker										
Hirano	1,742	JPY	0.2	16.5x	n.a.	-0.4%	-7%	6%	9%	
CKD	5,610	JPY	2.4	30.1x	15.6x	-1.1%	-9%	-15%	129%	
Wuxi Lead	34.2	CNY	8.4	22.8x	18.4x	2.7%	-2%	4%	24%	
Shenzhen Yinghe	20.1	CNY	1.9	21.9x	11.7x	1.1%	-1%	10%	-7%	
Zhejiang HangKe	22.9	CNY	2.1	23.4x	17.6x	0.2%	5%	8%	8%	
Guangdong Lyric	35.4	CNY	0.9	38.3x	15.6x	2.2%	-3%	5%	-43%	
Shenzhen United	19.7	CNY	1.0	20.4x	14.0x	1.1%	-3%	8%	-14%	

Source: Bloomberg (all estimates), Bernstein analysis
GLOBAL ENERGY STORAGE

Company Models

[CATL](#) | [LG Energy Solution](#) | [Samsung SDI](#)

[LG Chem](#) | [EcoPro BM](#) | [Posco Future M](#)

[Tianqi Lithium](#)

[Sungrow](#)

Industry Models

[Model: ESS tracker](#)

[Model: Monthly EV Tracker](#)

[Model: Battery Gigafactory Database](#)

[Industry model: Global EV TAM](#)

[Industry model: Global ESS TAM](#)

[Industry model: Battery TAM](#)

[Industry model: Battery supply and demand](#)

Blackbook

[Battery Utopia](#)

[Battery Tech Roadmap 2023: Faster, Higher, Stronger](#)

[Turn Up the Voltage... Investing Across the Battery Value Chain](#)

[Global Energy Storage: Resistance is Futile](#)

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[Global Energy Storage: Initiating Coverage. Lithium Positive \(Tianqi\) and Negative Cathodes \(EcoPro BM, Posco Future M\)](#)

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Recent Research

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[CATL: Initial perspectives on HK IPO Plans](#)

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[Global Energy Storage: Bigger and better batteries enables the rise of the 700km electric vehicle](#)

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BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	20 Aug 2026		TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
			Closing Price	Price Target		Cur	2025A	2026E	2027E	2025A	2026E	2027E
300750.CH (CATL)	O	CNY	385.00	800.00	7.6%	CNY	16.14	21.95	28.77	23.9	17.5	13.4
3750.HK (CATL)	O	HKD	629.00	770.00	24.4%	CNY	16.14	21.95	28.77	33.4	24.6	18.7
051910.KS (LG Chem)	M	KRW	264,500	312,000	(41.6)%	KRW (13,258.70)	2,693.22	28,743	(19.9)	98.2	9.2	
373220.KS (LGES)	M	KRW	358,500	351,000	(35.6)%	KRW (4,580.94)	(2,364.33)	10,225	(78.3)	(151.6)	35.1	
006400.KS (SDI)	M	KRW	498,500	520,000	99.4%	KRW (9,933.80)	1,751.66	18,577	(50.2)	284.6	26.8	
247540.KS (EcoPro BM)	U	KRW	113,300	140,000	(44.6)%	KRW	403.00	854.00	1,963.00	281.1	132.7	57.7
300274.CH (Sungrow)	O	RMB	110.89	185.00	(12.0)%	RMB	6.55	8.22	9.33	16.9	13.5	11.9
002466.CH (Tianqi Lithium)	O	CNY	46.03	80.00	(23.9)%	CNY	0.28	3.90	7.75	163.1	11.8	5.9
9696.HK (Tianqi Lithium)	O	HKD	33.74	65.00	(50.8)%	CNY	0.28	3.90	7.75	102.5	7.4	3.7
ASIA			1,976.42									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

DISCLOSURE APPENDIX

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